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Naive Bayes Classifier

$C_1, C_2, \ldots, C_K$ ← classes

$\vec{x}$: input (p-dim)

$\hat{y}$: to find

Bayesian principle :

Bayes theorem : $P(C_k \mid x_1 = x_1', x_2 = x_2', \ldots, x_p = x_p') = P(C_k \mid \vec{x}) = \frac{P(\vec{x} \mid C_k) \cdot P(C_k)}{P(\vec{x})}$

posterior ↓ likelihood ↓ prior ← evidence

Consider a spam filtering dataset with 2 classes: spam (S) and non-spam (N). Assume that the following probabilities are given.
Prior probabilities: P(S) = 0.2. P(N) = 0.8.
{ based on historical data
Likelihoods: P('free' | S) = 0.5, P('free' | N) = 0.1.
{ based on the occurence of word "free" in the historical data
Given a new email that contains the word 'free', will you classify it as spam?

→ $P(\text{spam} \mid \text{'free'}) = \frac{P(\text{'free'} \mid \text{spam}) \cdot P(\text{spam})}{P(\text{free})}$

$= \frac{(0.5)(0.2)}{(0.5)(0.2) + (0.1)(0.8)} = 0.5555 > 0.50$

Spam!

$P(S \mid \text{free}) = \frac{P(\text{free} \mid \text{spam}) \cdot P(\text{spam})}{P(\text{free})}$ $\qquad$ $P(N \mid \text{free}) = \frac{P(\text{free} \mid N) \cdot P(N)}{P(\text{free})}$

← denominators are same, and we need to compare, so we can ignore them in calculation. →

In higher dimensions

x_1 (A/B/C)	x_2 (X/Y)	x_3 (P/Q/R)	y (0/1)
A	Y	Q	1
⋮	⋮	⋮	⋮

10 points

$K = 2$, $p = 3$, $N = 10$; x_1: 3, x_2: 2, x_3: 3 diff. values

Figure out the class for $\vec{x} = (x_1 = B, x_2 = X, x_3 = R)$

$P(0 \mid x) = \frac{P(x \mid 0) \cdot P(0)}{P(x)}$ ← ignore

$P(0 \mid x) \propto P(x \mid 0) \cdot P(0)$

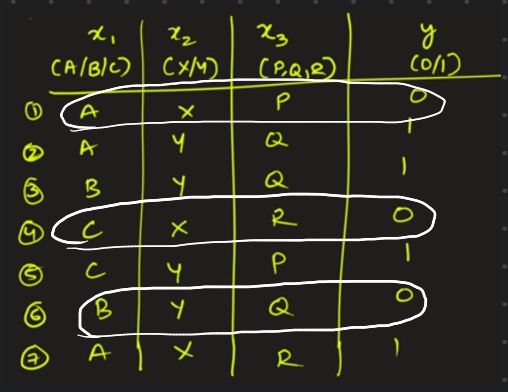
$P(0 \mid [B, X, R]) \propto P(x = [B, X, R] \mid 0) \cdot P(0)$

↳ How? $\qquad$ easy to calculate from the dataset

$P(x = [B, X, R] \mid 0) = P(x_1 = B, x_2 = X, x_3 = R \mid 0)$

(Naive assumption : conditional independence of input features given the label)

$= P(x_1 = B \mid 0) \cdot P(x_2 = X \mid 0) \cdot P(x_3 = R \mid 0)$



	x_1 (A/B/C)	x_2 (X/Y)	x_3 (P,Q,R)	y (0/1)
①	A	X	P	0
②	A	Y	Q	1
③	B	Y	Q	1
④	C	X	R	0
⑤	C	Y	P	1
⑥	B	Y	Q	0
⑦	A	X	R	1

$x_{new} = (x_1 = B, x_2 = X, x_3 = R)$

$P(y=0) = 3/7$

$P(y=1) = 4/7$

using the Naive Bayes assumption

$$P(y=0 \mid [B,X,R]) \propto P(x_1=B \mid 0) \cdot P(x_2=X \mid 0) \cdot P(x_3=R \mid 0) \cdot P(0)$$

$$= \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{3}{7}$$

$$= \frac{2}{63} \approx 0.0317$$

$$P(y=1 \mid [B,X,R]) \propto P(B \mid 1) \cdot P(X \mid 1) \cdot P(R \mid 1) \cdot P(1)$$

$$= \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{4}{7}$$

$$= \frac{1}{112} \approx 0.0089$$

$$P(y=0 \mid [B,X,R]) > P(y=1 \mid [B,X,R])$$

class 0!

The naive assumptions:

- The fundamental naive bayes assumption is that all predictors are conditionally independent given the class label. This assumption simplifies the calculation of likelihood P(x | Ck) by allowing it to be expressed as the product of probabilities of individual predictors.

$$P(x \mid C_k) = P(x_1, x_2, \dots, x_p \mid C_k) = P(x_1 \mid C_k) \cdot P(x_2 \mid C_k) \dots P(x_p \mid C_k)$$

$$P(x_1, \dots, x_p \mid C_k) = \prod_{i=1}^{p} P(x_i \mid C_k)$$

Advantages (of the assumption):

- Computational simplicity.
- Model training and scaling.
- Performance in the presence of irrelevant features.

Disadvantages:

- Feature correlation.
- Data scarcity and zero probability problem.
- Bias towards more frequent classes.

of parameters and the impact of the naive assumption

say each of the p features takes m values → m^p values for $\vec{x}$.

Using the Bayesian classifier:

$$P(C_K|x) \propto P(x|C_K) \cdot P(C_K)$$

For a particular C_K,

how many parameters do we need to compute? $(m^p - 1)$ likelihood terms [the last one is 1 − rest]

Doing this for all K classes,

$$\#\text{ of parameters for likelihood values} = K(m^p - 1)$$

For priors, $P(C_1), P(C_2), \ldots, P(C_K)$. we need to calculate $(K-1)$ of these.

$$\text{Total \# of parameters to compute using the Bayesian classifier} = \underbrace{K(m^p - 1)}_{\text{likelihood}} + \underbrace{(K-1)}_{\text{prior}}$$

using the Naive Bayes classifier:

$$P(C_K|x) \propto P(x_1|C_K) \cdot P(x_2|C_K) \cdots P(x_p|C_K)$$

each x_i takes m values. And $\{x_i\}_{i=1}^{p}$ → $(m-1)p$ likelihood terms

For K classes → $Kp(m-1)$ likelihood terms

Total # of parameters → $\underbrace{Kp(m-1)}_{\text{likelihoods}} + \underbrace{(K-1)}_{\text{priors}}$

e.g. 10 features, each can take 5 values. 3 classes.

→ with Bayes classifier → ≈ 30 million

with Naive bayes → ≈ 122 WTF!

Q.20	Given a dataset with K binary-valued attributes (where $K > 2$) for a two-class classification task, the number of parameters to be estimated for learning a naïve Bayes classifier is
(A)	$2^K + 1$
(B)	$2K + 1$
(C)	$2^{K+1} + 1$
(D)	$K^2 + 1$

K attributes, each can take 2 values

$P(x_1|0) \cdot P(x_2|0) \cdots P(x_K|0)$

binary valued

$2 + 2 + \ldots + 2 = (2K)$ } $2K$

But we can leave out 1 so

$(2-1)K = K$

for 1 class → K

since we have 2 classes → $2K$.

$(2-1)k \cdot 2 + (2-1)$

each feature takes 2 values; # of features; # of classes; # of classes − 1

prior terms → 1 (calc for $y=0$, deduce for $y=1$)

$(2K+1)$

Algo

Bayes formula: $P(C_k|\vec{x}) = \frac{P(\vec{x}|C_k) \cdot P(C_k)}{P(\vec{x})}$

Naive Bayes assumption: $P(x_1, ..., x_p|C_k) = P(x_1|C_k) \cdot \ldots \cdot P(x_p|C_k)$.

1. Calculate prior probabilities using the dataset: $P(y = C_1), \ldots, P(y = C_k)$.
2. Calculate the likelihoods: $P(x_1|y = C_1), \ldots, P(x_p|y = C_1)$.
3. Calculate the posterior probability for all classes: $P(C_1|x), \ldots, P(C_k|x)$.
4. Assign the label with the highest probability.

Q: using the naive Bayes classifier:

Candidate	Skill Level	Certification	Previous Job Success	Education Relevance	Hired
1	High	Yes	High	Relevant	Yes
2	High	No	Low	Irrelevant	No
3	Medium	Yes	Medium	Relevant	Yes
4	Low	No	Low	Irrelevant	No
5	Medium	No	Medium	Irrelevant	No
6	Low	Yes	High	Relevant	No
7	High	Yes	Medium	Irrelevant	Yes
8	Medium	Yes	Low	Relevant	No
9	High	No	High	Relevant	Yes
10	Low	No	Medium	Irrelevant	No

classify (medium, yes, high, relevant).

$$P(\text{hired} \mid \langle \text{medium, yes, high, relevant} \rangle) = P(\text{medium}|\text{hired}) \cdot P(\text{yes}|\text{hired}) \cdot P(\text{high}|\text{hired}) \cdot P(\text{relevant}|\text{hired}) \cdot P(\text{hired})$$

$$= \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{2}{\cancel{9}_2} \cdot \frac{3}{\cancel{4}} \cdot \frac{\cancel{4}}{10}$$

$$= \frac{9}{320} = 0.028$$

$$P(\overline{\text{hired}} \mid \langle \text{med, yes, high, relevant} \rangle) = \frac{2}{6} \cdot \frac{2}{6} \cdot \frac{1}{6} \cdot \frac{2}{6} \cdot \frac{6}{10} = \frac{1}{270} = 0.0037$$

$\Rightarrow$ person is hired.

Laplace smoothing

Candidate	Skill Level	Certification	Previous Job Success	Education Relevance	Hired
1	High	Yes	High	Relevant	Yes
2	High	No	Low	Irrelevant	No
3	Low	Yes	Medium	Relevant	Yes
4	Low	No	Low	Irrelevant	No
5	Medium	No	Medium	Irrelevant	No
6	Low	Yes	~~High~~ medium	Relevant	No
7	High	Yes	Medium	Irrelevant	Yes
8	Medium	Yes	Low	Relevant	No
9	High	No	High	Relevant	Yes
10	Low	No	Medium	Irrelevant	No

say there's no medium instance for skill level that is hired.
Then $P(\text{skill}=\text{medium} \mid \text{yes}) = 0$

Ily, there's no high instance of prev. job success for hired = no.
then $P(\text{prev.job suc} = \text{high} \mid \text{hired} = \text{no}) = 0$

Then, $P(\text{hired} = \text{yes} \mid \langle \text{medium, yes, high, relevant} \rangle) = 0$
$P(\text{no} \mid \langle \text{medium, yes, high, relevant} \rangle) = 0$
} we are unable to make any conclusion.

Laplace smoothing deals with this zero probability effect.

Without Laplace smoothing:

$$P(\text{feature} \mid \text{class}) = \frac{\#\text{ of times the feature appears in class}}{\text{total \# of instances in class}}$$

With Laplace smoothing:

$$P(\text{feature} \mid \text{class}) = \frac{\#\text{ of times the feature appears in class} + \alpha}{\text{total \# of instances in class} + m}$$

m: # of different values this feature can take

αm

e.g.

(A/B/C) (P/Q)

X_1	X_2	Y
A	P	0
B	P	1
C	Q	0
A	P	0

Without Laplace smoothing:

$P(X_1 = B \mid Y = 0) = \frac{0}{3} = 0$

$P(X_1 = A \mid Y = 0) = \frac{2}{3}$

$P(X_1 = C \mid Y = 0) = 1/3$

With Laplace smoothing:

$P(X_1 = B \mid Y = 0) = \frac{0+1}{3+3} = \frac{1}{6}$

X_1 takes values $\{A, B, C\}$ 3 values

$P(X_1 = A \mid Y = 0) = \frac{2+1}{3+3} = \frac{3}{6}$

$P(X_1 = C \mid Y = 0) = \frac{1+1}{3+3} = \frac{2}{6}$

In the denominator, we add m, so that the probabilities sum up to 1.

$\frac{c_1}{c} + \frac{c_2}{c} + \cdots + \frac{c_m}{c} = 1$
$\Rightarrow c_1 + \cdots + c_m = c$

$$\frac{c_1+1}{c+m} + \frac{c_2+1}{c+m} + \cdots + \frac{c_m+1}{c+m} = \frac{c_1 + \cdots + c_m + m}{c+m} = \frac{c+m}{c+m} = 1$$

say $x_{new} = \langle B, Q \rangle$

Without Laplace smoothing:

$P(0 \mid \langle B, Q \rangle) \propto P(B \mid 0) \cdot P(Q \mid 0) \cdot P(0)$
$= 0 \cdot \frac{1}{3} \cdot \frac{3}{4} = 0$

$P(1 \mid \langle B, Q \rangle) \propto P(B \mid 1) \cdot P(Q \mid 1) \cdot P(1)$
$= \frac{1}{1} \cdot \frac{0}{1} \cdot \frac{1}{4} = 0$

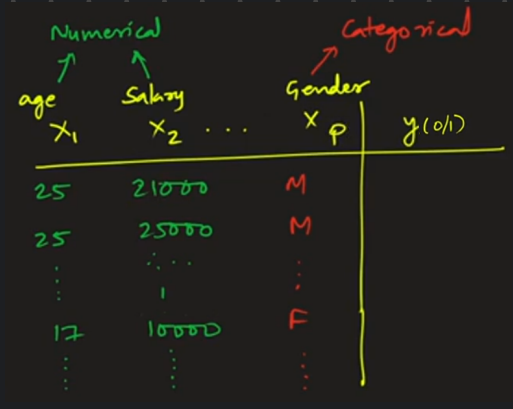
With Laplace smoothing:

$P(0 \mid \langle B, Q \rangle) \propto P(B \mid 0) \cdot P(Q \mid 0) \cdot P(0)$
$= \frac{0+1}{3+3} \cdot \frac{1+1}{3+2} \cdot \frac{3}{4} = \frac{1}{6} \cdot \frac{2}{5} \cdot \frac{3}{4} = \frac{1}{20} = 0.05$

$P(1 \mid \langle B, Q \rangle) \propto P(B \mid 1) \cdot P(Q \mid 1) \cdot P(1)$
$= \frac{1+1}{1+3} \cdot \frac{0+1}{1+2} \cdot \frac{1}{4} = \frac{2}{4} \cdot \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{24} = 0.0416$

$\Rightarrow \hat{y}_{new} = 0$

Handling numerical data in naive Bayes classifiers



Ways:

① Binning the values → turns them into categorical variables

② Assume a distribution for the values.

Assume $P(X_1 \mid y=0)$ has some distribution.

Then calculate the parameters of the distribution using MLE.

Then calculate $P(x_1 = 25 \mid y=0)$ using the distribution.

NOTE: Laplace smoothing won't be able to solve this problem. # of values the features can take is unknown.

MLE: try to maximize log-likelihood function.

Gaussian naive Bayes

Age (x_1)	Salary (x_2)	Education (x_3)	Buy (c)
25	50,000	bachelor's	not buy
30	80,000	masters	buy
35	65,000	high school	not buy
40	100,000	bachelor's	buy

Non-categorical data.

$x_{new} = \langle 32, 75000, \text{Bachelor's} \rangle$

Assume normal distribution:

$\text{Age}, \text{Buy} \sim N(\mu_{A,B}, \sigma^2_{A,B})$

$\text{Age}, \text{NA Buy} \sim N(\mu_{A,NB}, \sigma^2_{A,NB})$

$\text{Sal}, \text{Buy} \sim N(\mu_{S,B}, \sigma^2_{S,B})$

$S, NB \sim N(\mu_{S,NB}, \sigma^2_{S,NB})$

$\mu_{A,B} = \frac{30+40}{2} = 35 \qquad \sigma^2_{A,B} = \frac{(30-35)^2 + (40-35)^2}{2} = 5^2$

$\text{Age}, \text{Buy} \sim N(\mu = 35, \sigma^2 = 25)$

$$P(\text{Age} = 32 \mid \text{Buy}) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{32-35}{5}\right)^2} \approx 0.067$$

Point probability for continuous distribution is zero. So this is incorrect. But prob. is $\propto$ this value, so it's a decent enough estimate. Like, I don't care as long as I can make correct comparisons.

Bernoulli naive Bayes

Tweet id	Contains "Happy" (0/1)	Contains "Sad" (0/1)	Positive Sentiment
1	1	0	1
2	0	1	0
3	1	1	1

$x_{new} = (1, 0)$

NOTE: we can do this using normal Naive Bayes (/Laplace smoothing) as well.

$\text{Happy} = 1, y = 1 \sim \text{Bernoulli}(p) \quad \{1, 1\} \quad p = \frac{2}{2} = 1$

$\text{Happy} = 0, y = 1 \sim \text{Bernoulli}(p) \quad p = \frac{0}{2} = 0$

$\text{Sad} = 1, y = 1 \sim \text{Bernoulli}(p) \quad \{0, 1\} \quad p = \frac{1}{2}$

$\text{Sad} = 0, y = 1 \sim \text{Bernoulli}(p) \quad p = \frac{1}{2}$

$$P(\langle 1, 0 \rangle \mid y=1) \propto P(\text{Happy}=1 \mid y=1) \cdot P(\text{Sad}=1 \mid y=1) \cdot P(y=1) = 1 \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$$

Same as categorical data.

Multinomial naive Bayes

$$P(x_1 = n_1, x_2 = n_2, \ldots, x_p = n_p \mid y = c) \propto \theta_{1,c}^{n_1} \cdot \theta_{2,c}^{n_2} \cdot \theta_{3,c}^{n_3} \cdots \theta_{p,c}^{n_p}$$

$= \frac{n!}{x_1! x_2! \ldots x_p!} \left(\theta_{1,c}^{x_1}\right)\left(\theta_{2,c}^{x_2}\right)\cdots$ ← ignore (same for all terms)

whenever we've got counts of stuff (e.g. # of times a word appears in a tweet), the multinomial distribn" works really well.

e.g.

Article id	n_1 "win"	n_2 "game"	n_3 "computer"	n_4 "election"	Category
1	3	2	0	0	Sports
2	0	0	2	1	Techology
3	0	1	1	2	Politics
4	1	2	0	0	Sports
5	0	0	3	0	Technology

$n_{new} = \langle 1,0,1,1 \rangle$

use multinomial NB + Laplace smoothing to predict $\hat{y}_{new}$.

$$P(\langle 1,0,1,1 \rangle \mid y = \text{sports}) \propto (\theta_{n_1,\text{sports}})^1 \cdot (\theta_{n_2,\text{sports}})^0 \cdot (\theta_{n_3,\text{sports}})^1 (\theta_{n_4,\text{sports}})^1$$

$$\theta_{x_i, y=c} = \frac{\text{count}(x_i \mid y=c)}{\sum_j \text{count}(x_j \mid y=c)} = \frac{\text{count}(x_i \mid y=c) + 1}{\sum_j [\text{count}(x_j \mid y=c) + 1]}$$

$$\theta_{n_1,\text{sports}} = \frac{(3+1)+1}{\underbrace{(3+1)}_{n_1|\text{sports}} + \underbrace{(2+1)}_{n_2|\text{sports}} + \underbrace{(0+1)}_{n_3|\text{sports}} + \underbrace{(0+1)}_{n_4|\text{sports}}} = \frac{4+1}{8+4} = \frac{5}{12}$$

$$\theta_{n_2,\text{sports}} = \frac{(2+2)+1}{8+4} = \frac{5}{12}$$

$$\theta_{n_3,\text{sports}} = \frac{(0+0)+1}{8+4} = \frac{1}{12}$$

$$\theta_{n_4,\text{sports}} = \frac{(0+0)+1}{8+4} = \frac{1}{12}$$

$$P(\langle 1,0,1,1 \rangle \mid y = \text{sports}) \propto \left(\frac{5}{12}\right)^1 \left(\frac{5}{12}\right)^0 \left(\frac{1}{12}\right)^1 \left(\frac{1}{12}\right)^1 = \left(\frac{5}{12^3}\right)$$

$$P(y = \text{sports} \mid n = \langle 1,0,1,1 \rangle) \propto P(n = \langle 1,0,1,1 \rangle \mid y = \text{sports}) \cdot P(y = \text{sports})$$

$$= \frac{5}{12^3} \cdot \frac{2}{5} = \frac{2}{12^3} \approx 0.0012$$

Ily,

$$\theta_{x_1,\text{Tech}} = \frac{0+1}{6+4} = \frac{1}{10}, \quad \theta_{x_2,\text{Tech}} = \frac{0+1}{6+4} = \frac{1}{10}, \quad \theta_{x_3,\text{Tech}} = \frac{5+1}{6+4} = \frac{6}{10},$$

$$\theta_{x_4,\text{Tech}} = \frac{1+1}{6+4} = \frac{2}{10}$$

$$P(y = \text{Tech} \mid x = 1,0,1,1) \propto P(x = 1,0,1,1 \mid y = \text{Tech}) \cdot P(y = \text{Tech})$$

$$\propto \left(\frac{1}{10}\right)^1 \cdot \left(\frac{1}{10}\right)^0 \cdot \left(\frac{6}{10}\right)^1 \cdot \left(\frac{2}{10}\right)^1 \cdot \frac{2}{5}$$

$$= \frac{24}{5000} = 0.0048$$

$$\theta_{x_1,\text{Politics}} = \frac{0+1}{4+4} = \frac{1}{8}, \quad \theta_{x_2,\text{Politics}} = \frac{1+1}{4+4} = \frac{2}{8}, \quad \theta_{x_3,\text{Politics}} = \frac{1+1}{4+4} = \frac{2}{8}$$

$$\theta_{x_4,\text{Politics}} = \frac{2+1}{4+4} = \frac{3}{8}$$

$$P(y = \text{politics} \mid x = 1,0,1,1) \propto \left(\frac{1}{8}\right)^1 \cdot \left(\frac{2}{8}\right)^0 \cdot \left(\frac{2}{8}\right)^1 \cdot \left(\frac{3}{8}\right)^1 \cdot \frac{1}{5}$$

$$= \frac{6}{8^3 \times 5} \simeq 0.0023$$

$\Rightarrow \hat{y}_{new} = \text{technology}$